Do bridge-monitoring error bars survive an unseen season? Physics-guided, conformally calibrated temperature compensation on the KW51 railway bridge

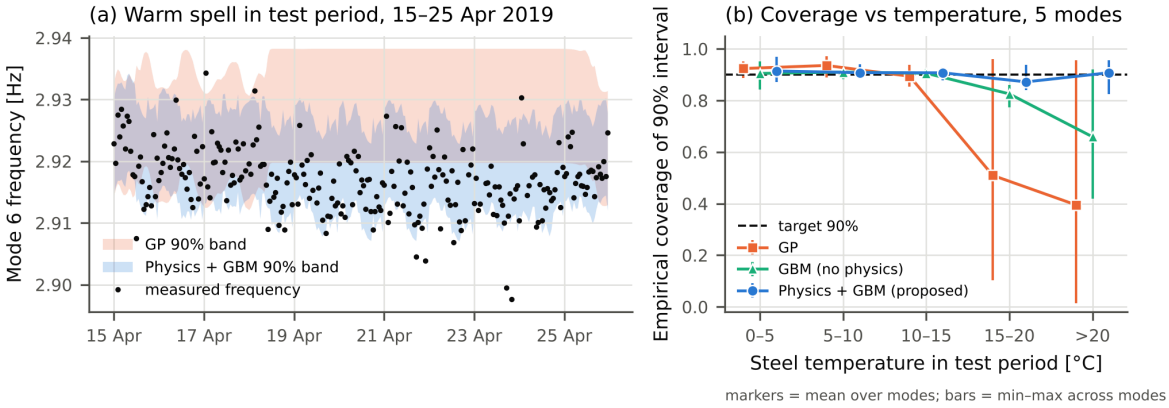
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PROBLEM

Structural health monitoring tracks a bridge's natural frequencies, because damage lowers stiffness and therefore frequency. Temperature changes frequency by a similar amount, so the measurement must be *compensated*, and the alarm threshold is only as trustworthy as its uncertainty band. Conformal prediction gives bands with a coverage guarantee, but only if past and future data are exchangeable. Seasonal data are not. **How much does coverage degrade when a model must predict a season it has not seen, and does a physics prior help?**

DATA AND METHOD

I used the public KW51 dataset (Maes & Lombaert, Zenodo): hourly natural frequencies of a steel railway bridge in Leuven, Belgium, Oct 2018–Jan 2020, retrofitted in May–Sep 2019. No new experiments were run. I trained on Nov–Jan, calibrated on Feb, and tested on Mar–mid-May, when temperatures reached 26 °C against a training maximum of 13.6 °C. The proposed model is a 4-parameter thermo-elastic curve, $f = f_0\sqrt{(1+a\Delta T)}$ plus a cold-stiffening term, with a small gradient-boosted model fitted to its residuals. It was compared with linear, cubic, Gaussian-process and physics-free boosted-tree baselines, all wrapped in the same split-conformal procedure. Five vibration modes were analysed, with 9 models including ablations and 5 seasonal folds.



Key figure. (a) Mode 6 during a warm spell in the test period: the Gaussian-process 90 % band drifts away from the measurements, while the physics-guided band follows them. (b) Empirical coverage of nominal 90 % intervals by test temperature, averaged over 5 modes (bars: min–max across modes). Training never exceeded 13.6 °C.

RESULTS SO FAR

- With a random (exchangeable) split, all 9 models were calibrated (0.90 ± 0.01), as theory predicts.
- On test hours warmer than anything in training (2 seasonal folds × 5 modes), the physics-guided model covered a median of **0.90 (worst 0.84)**. Physics-free boosted trees covered 0.77 (worst 0.59), a GP with a trend kernel 0.69, and a standard GP 0.36.
- Ablations show *why*: the cold-stiffening term keeps the freeze-induced frequency jump out of the warm-weather slope. Without it, the fitted slope is up to about 2× too steep and coverage drops to 0.75. Adding a plain linear trend did not help.
- **Negative result:** trained on spring and tested on winter, every model, including mine, failed below 0 °C. Physics cannot extrapolate into a mechanism the data never showed.
- Frozen bands flagged the 2019 retrofit on every post-retrofit day for 3 modes, with 0–1.3 % false-alarm days.

LIMITATIONS AND NEXT STEPS

This is one bridge, and out-of-range samples are small (90–379 h per fold). The retrofit was a large stiffening, not damage, so a minimum-detectable-shift study is needed. Next: a cross-season prior for the cold term, a second structure (LUMO), and a physics-informed neural-network residual. **I would value feedback on the physical cause of the cold stiffening and on evaluation design.**

All numbers are reproducible from the project repository. Prior work on KW51 (Maes et al. 2022; Wittenberg et al. 2025; Neumann et al. 2025–26) covers environmental compensation and monitoring. In my search I did not find conformal coverage evaluated under seasonal shift for this dataset.